Why Expertise Creates Blind Spots

The Azubi Method, Part 1

The ELIZA Effect, Sycophancy, Operational Blindness, and Three Error Classes

The assumption behind most quality assurance systems is: the more expertise, the better the results. This assumption is half right and half dangerous.

Expertise increases depth. A specialist finds problems in their domain that a generalist overlooks. But expertise simultaneously reduces breadth. Anyone who has deeply mastered a field develops routines. Routines save time and increase efficiency. But routines create patterns. And patterns create blind spots. What lies within the routine is no longer examined. What is no longer examined becomes an assumption. What becomes an assumption becomes invisible.

This is not individual failure. It is a property of expertise itself.

In human teams, this effect is mitigated by two natural mechanisms: turnover and individuality. New team members bring fresh perspectives. And even long-standing colleagues have different experiences, different training, different cognitive patterns.

With AI agents, both mechanisms disappear. And three additional mechanisms exacerbate the problem.

The ELIZA Effect

In 1966, Joseph Weizenbaum discovered that humans attribute more capability to machines than they actually possess. His program ELIZA understood nothing. But users attributed comprehension to it because the surface was convincing. With modern language models, this cognitive tendency has not weakened but strengthened, because the surface deceives far more effectively.

AI-generated results almost always look plausible. This lowers the willingness to scrutinize. Studies show that users of AI tools systematically overestimate their own speed and quality. The plausibility of the output is confused with correctness.

Sycophancy

Sycophancy describes the tendency of language models to agree with the user rather than contradict them. This is not a bug. It is a direct consequence of training: models are optimized through human feedback, and humans systematically rate answers that agree with them higher than answers that contradict them.

Anyone who asks an AI model to evaluate a result will feel this tendency: downplaying problems, amplifying praise, avoiding contradictions. The result: the user receives the confirmation they unconsciously seek, instead of the contradiction they need.

Current research shows the problem is a spectrum. The Lechmazur Sycophancy Benchmark (2026) measures not only sycophancy but also its counterpart: compulsive contradiction. Models adjusted through anti-sycophancy training learned to be categorically skeptical rather than situationally discerning. Sycophancy and compulsive criticism are not two different problems. They are two poles of the same problem: a lack of situational authenticity. The Azubi Method bypasses both poles because it neither agrees nor disagrees—it asks.

Operational Blindness in AI Agents

An agent that has been configured once stays configured. It does not get tired. But it does not get curious either. It develops no new perspectives, asks no unexpected questions, brings no experience from another project. On day 100, it is the same agent as on day 1.

When two agents are based on the same model, they share the same training and the same bias structures. Different system prompts create different focus areas. But not different thinking patterns. Five different roles on the same model are five different lenses on the same eyes. The lenses change the focus. The eyes remain the same.

The Triad

These three mechanisms work together: The ELIZA effect lowers the willingness to scrutinize. Sycophancy confirms one’s own assumptions. Operational blindness prevents noticing that both are happening. The result is a systematic review gap: results that look plausible, are confirmed by the AI, and are questioned by no one involved. Not because everyone is incompetent. But because the mechanisms work as designed.

Three Error Classes

In practice, three error classes have crystallized, each requiring different discovery methods:

Known patterns. Cataloged errors that can be covered by automated tests. Rule-based checks find them reliably. No expert is needed for these.

Expected logic errors. Errors that require domain knowledge. An experienced reviewer recognizes them because they know where they typically occur. Here, expertise is indispensable.

Silent assumptions. Errors that produce no error message, no warning, no visible symptom. The system continues running, apparently correct. A value silently truncated. A process running into the void since system startup. A return value overwritten by a subsequent operation.

The third class is the most dangerous. And no expert finds it. Not because they are incompetent, but because it lies below their attention threshold. They search for complex problems. The problem is simple. So simple that it becomes invisible.

The Attention Threshold

There is a boundary below which experts stop searching. Everything above this threshold is examined: the complex, the known, the expected. Everything below is ignored: the simple, the obvious, the self-evident. The most dangerous errors lie below this threshold.

This threshold exists for human experts and AI agents alike. An expert prompt produces expert behavior: deep analysis within the defined domain. And precisely because of that: systematic blind spots. The irony: the better the expert prompt is formulated, the more reliably it reproduces the operational blindness it was supposed to uncover.

Three Layers, Three Discovery Profiles

The three error classes require three different discovery methods. No single method covers all three.

Layer 1: Automated testing. Finds known error classes. Pattern matching against defined rules. Fast, reproducible, exhaustive within the catalog. Blind to everything outside it.

Layer 2: Expert review. Finds expected logic errors. Knowledge-based analysis. Deep within the domain. Blind to the obvious.

Layer 3: Context-free instance. Finds silent assumptions. Naive questions without prior knowledge. No domain, no context, no memory. Sees the obvious because it knows nothing else.

In practice, the most critical findings were distributed roughly equally across all three layers. The third layer required the least effort. But had the highest hit rate.

Knowledge creates blind spots. Ignorance creates questions. And questions find more than answers.